MATHEMATICS

Least squares solution for $y = ax^b$ after logarithmic transformation

Asked 9 years, 10 months ago Modified 6 years, 8 months ago Viewed 3k times

0

How can I linearize a nonlinear model using

$$y = Ax^b$$

and

$$y = A \ln x + B$$

I couldn't find much online using the above methods.

The notes are taken from a class in Models in Applied Mathematics

linear-algebra mathematical-modeling Edit tags

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edited Apr 19, 2017 at 4:53
dantopa
10.3k 10 44 82

asked Feb 10, 2014 at 7:29
iordanis
813 3 10 14

3 Answers

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Problem statement

Given a sequence of $m = 5$ measurements $\{x_k, y_k\}_{k=1}^m$ and the model

$$y(x) = ax^b.$$

Use the method of least squares to find the solution

$$\boxed{(a,b)_{LS} = \left\{ (a,b) \in \mathbb{R}^2 \colon \sum_{k=1}^{m} \left(y_k - ax_k^b\right)^2 \text{ is minimized} \right\}} \tag{1}$$

The target of minimization is the merit function ,

$$M(a,b) = \sum_{k=1}^{m} \left(y_k - ax_k^b\right)^2 \tag{2}$$

Defining the residual error as,

$$r_k = y_k - ax_k^b$$

the least squares problem minimizes the sum of the squares of the residual errors $r \cdot r = r^2$

Data

The input parameters are $(a,b) = (1, \frac{1}{2})$, and random noise $|\epsilon| < 0.1$ is stirred in. The data set in then

k	x	y
1	0.2	0.510691
2	0.4	0.554739
3	0.6	0.832502
4	0.8	0.831988
5	1	0.948272

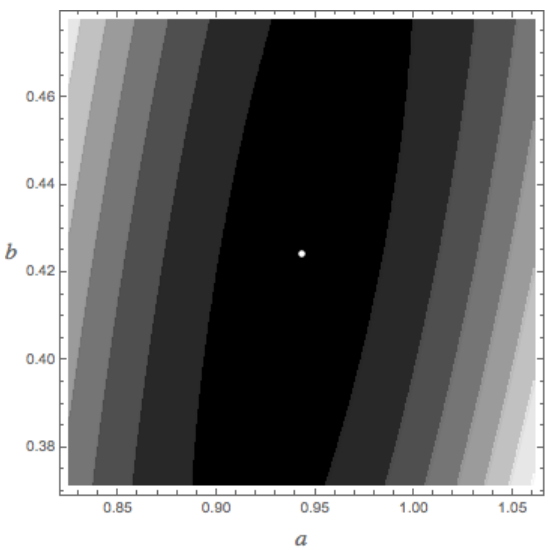
Least squares solution

The least squares solution described in (1) is

$$(a,b)_{LS} = (0.943281, 0.424359)$$

with total error of $r^2 = 0.0143$

The solution point is displayed against a contour plot of the merit function in (1). The least squares solution is nestled in at the very least of the total error.



Least squares solution of reduced system

What if we don't have access to a conjugate gradient code? The problem can be simplified using least squares. The considerable challenge of finding a minimum in two dimensions will be supplanted with a minimization problem in one dimension.

Go back to equation (1), and attack with the conventional tools from the calculus. Demand the derivative with respect to a be 0

$$\frac{\partial}{\partial a}\sum_{k=1}^{m}\left(y_k - a x_k^b\right)^2 = -2\sum_{k=1}^{m}\left(y_k - a x_k^b\right) x_k^b = 0$$

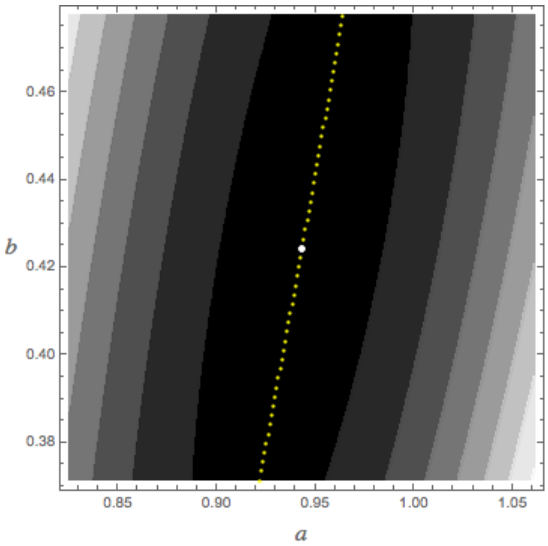
This produces a constrained version of the parameter

$$a_*(b) = \frac{\sum y_k x_k^b}{\sum x_k^{2b}}$$

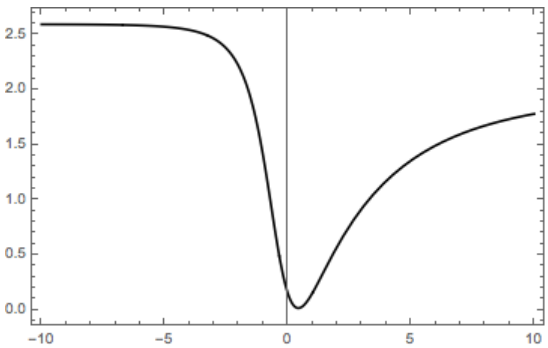
Given b_{LS}, we can compute a_{LS} like so

$$a_{LS} = a_*(b_{LS}) \tag{3}$$

The figure below plots $a_*(b)$ from (3) as a dashed line against the full two dimensional merit function.



Instead of searching in two dimensions, the problem involves searching along this trajectory.



The constrained merit function is

$$M_*(b) = M\left(a_*(b), b\right) = \sum_{k=1}^{m} \left(y_k - a_* x_k^b\right)^2 = \sum_{k=1}^{m} \left(y_k - \frac{\sum y_k x_k^b}{\sum x_k^{2b}} x_k^b\right)^2 \tag{4}$$

The function is a delight to minimize. It has a lone minimum and is monotonic is the best way. Moving from left to right, the function decreases monotonically up to the minimum, and then increases monotonically.

Find b_{LS} by finding the minimum of the (4), use (3) to compute a_{LS} . This exactly matches the full least squares solution and minimum error.

Logarithmic transformation

A logarithmic transformation is used to batter the data set until in can be forced into a linear system. In the limit of no error,

$$\begin{aligned} y(x) &= ax^b \\ \ln y(x) &= \ln ax^b = \ln a + b \ln x \end{aligned}$$

is an exact solution. But if there were no error, there would be no need for least squares. A way to think about the problem is this: the method of least squares takes a "peanut butter" to smoothing out the errors. A residual error of 1 has the same effect whether $y = 1$ or $y = 1000$. The logarithm enforces a location dependence. If $y = 1$ and $r = 1$, the prediction is $y = 10$. The difference in our linear space is 9. If that same residual error is measured at $y = 1000$, the prediction is $10,000$, a difference of 9000. This is very disruptive for the solution.

Proceeding, the battered data is inserted in the linear system

$$\mathbf{A}\,\alpha = \tilde{y}$$

$$\begin{bmatrix} 1 & \ln x \end{bmatrix} \begin{bmatrix} \ln a \\ b \end{bmatrix} = \begin{bmatrix} \ln y \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1.60944 \\ 1 & -0.916291 \\ 1 & -0.510826 \\ 1 & -0.223144 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \ln a \\ b \end{bmatrix} = \begin{bmatrix} -0.67199 \\ -0.589257 \\ -0.18332 \\ -0.183937 \\ -0.0531137 \end{bmatrix}$$

The solution for the linear system is

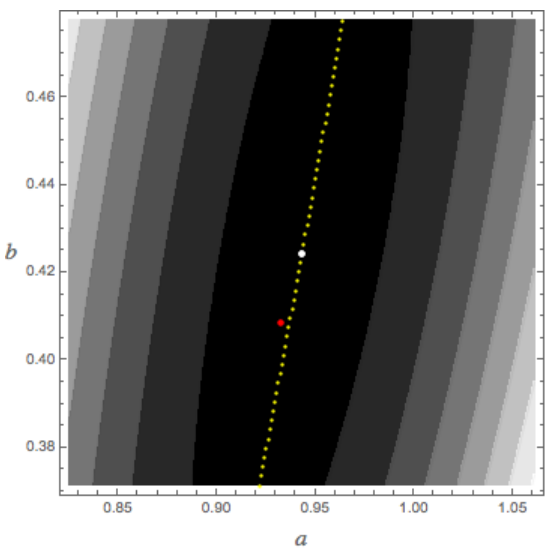
$$\mathbf{A}^{+}\alpha = \begin{bmatrix} \ln a \\ b \end{bmatrix} = \begin{bmatrix} -0.0700446 \\ 0.408441 \end{bmatrix}$$

To compare solutions

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0.932352 \\ 0.408441 \end{bmatrix}$$

The total error is $r^2 = 0.0146$.

How close are we? The red dot marks the solution found after logarithmic transformation.



In closing, a long post has been written which provides details for the insight of @Claude Leibovici:

> So, using the estimates obtained by the linear regression, you must start the nonlinear regression.

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edited Apr 19, 2017 at 17:15

answered Apr 18, 2017 at 23:54

dantopa
10.3k 10 44 82

3

Suppose you have to fit experimental data using a model such that $y = ax^b$ you should first notice that this model is nonlinear with respect to its parameters. The problem with nonlinear regression is that you must provide "reasonable" starting guesses for parameters a and b.

In some cases, you can linearize the model; in your case, taking logarithms of both sides, you have $\log(y) = \log(a) + b\log(x)$ which is of the form $Y = A + B\,X$, using $Y = \log(y)$, $A = log(a)$, $B = b$ and $X = \log(x)$. So, standard linear regression can be used from which parameters A and B will be obtained. So, going backwards, $b = B$ and $a = e^A$.

However, this is not the end of the story for the simple reason that, when you use the linearized form, it means that you minimize the sum of the squares of the errors for Y that is to say for $log(y)$; while the original problem is to minimize the sum of the squares of the errors for y which is not the same.

So, using the estimates obtained by the linear regression, you must start the nonlinear regression. For sure, in the case of marginal errors, the estimates will be quite close to the solution.

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answered Feb 10, 2014 at 8:40
Claude Leibovici
257k 52 112 231

So the point of linearizing a data set is so that it can be represented by a line? Or so that I can find the variables described above and fit it on a logistic model? – iordanis Feb 10, 2014 at 8:43

You are right but on a log-log plot. – Claude Leibovici Feb 10, 2014 at 8:44

I would appreciate it if you could link me to an example that makes use of the above technique. – iordanis Feb 10, 2014 at 8:47

I give you a few data points : (1,1.65), (2,4.91), (3,9.24), (4,14.6), (5,20.7), (6,27.4). Now, apply and send me your estimated values for the original parameters a and b. Cheers. – Claude Leibovici Feb 10, 2014 at 8:53

Solve the system of -2*e^B*(1.65-Ae^B)-2*4.91e^(2B)(1-A*e^(2B))-2*9.24*e^(3B)(1-Ae^(3B))=0 and d(Sum(y-Ae^(Bx))^2)/dB=0 and obtain A and B and then replace them on Y=A+BX? – iordanis Feb 10, 2014 at 9:08

1

Take logs

$$\ln(y) = \ln(A) + b\ln(x)$$

which we can define as

$$Y = A + BX.$$

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answered Feb 10, 2014 at 7:33
Chinny84
14.1k 2 22 31

I appreciate your quick response but can you demonstrate with an example? I would greatly appreciate it! – iordanis Feb 10, 2014 at 7:34

How do you mean? – Chinny84 Feb 10, 2014 at 7:35

An example that you linearize a nonlinear model. – iordanis Feb 10, 2014 at 7:37

Is this for a regression model? If so with the linearise equation you can use a simple linear regression to fit the data. – Chinny84 Feb 10, 2014 at 7:42

I am not sure exactly what it is for but these are the 2 equations I found on my notes. And I am just trying to study more about this subject but can't find resources. If you can identify it I would appreciate it. – iordanis Feb 10, 2014 at 7:44